LIMER: Link-Level Detection and Localized Recovery in Active-Active GPU Training Fabrics

Author One*
Institution
first.author@example.com

Abstract

Gray link failures preserve connectivity while impairing communication. LIMER studies timely path-level detection, cost-aware local rerouting and execution consistency in an active-active dual-plane fabric. We implement a packet-level RDMA simulation loop that combines periodic same-path canaries, actual retransmission and ACK progress, and receiver packet-spacing changes. A controller uses delivered alarms and measured three-leg helper routes to compare the cost of staying with coordinated replacement. Across four registered fault conditions, each selected detection cohort has 60/60 trials correctly localized strictly below 100 ms; the largest per-condition p95 is 43.12 ms and the largest observed latency is 47.63 ms. The 640 new runs pass execution audits and independent event replay; this count includes unfinished communication controls. In a matched slow-helper boundary, benefit gating reduces mean completion from 440.80 to 389.54 ms relative to direct replacement, but adds 5.75 and 11.60 ms under persistent blackholes and weak loss. Earlier validation failures and a corrected loss-stream sampling issue remain in the evidence. The results are conditional on healthy startup calibration, ideal endpoint observations and the registered workload. They do not establish universal gray-fault detection, sustained replacement-plane throughput, numerical AllReduce correctness or superiority over external recovery systems.

Keywords

GPU clusters, collective communication, fault tolerance, gray failures, datacenter networks, active-active fabrics

1 Introduction

An access link can remain UP while loss or severe delay slows a collective. Consider two network planes already carrying data: when plane B becomes unreliable, its unacknowledged data can delay dependent communication even if the scheduler sends new work over A. A useful response must identify the impaired path quickly, decide whether a helper route repays its setup cost, and preserve ownership while unfinished data changes route.

Our primary target is a correctly localized controller alarm strictly below 100 ms in at least 95% of trials in each registered fault condition. We detect impaired delivery, including severe slowdown; we do not require proof of permanent packet loss. Fast detection depends on the opportunity to observe the impairment. Passive progress can be sparse, while probes consume bandwidth and need not expose every fault seen by business traffic. A timing result is therefore meaningful only with its traffic, calibration, measurement and reporting assumptions stated.

LIMER now connects detection and recovery in one native packet simulation. A 64-KiB canary traverses every original sender/plane path every 2 ms. Distinct canary failures, actual retransmissions, business ACK stalls and a sustained shift in receiver packet spacing provide complementary evidence. An alarm then travels through the modeled control FIFO. The controller measures a three-leg route through an idle GPU in each endpoint server and compares the estimated completion cost of staying and replacing B. A versioned transaction and per-chunk acknowledgment prefix govern rescue of unfinished data. Both physical planes remain present.

We organize the study around three questions:

(1) **Fast detection:** can each registered fault condition meet the 95% below-100-ms target, and what observations and overhead does that require?
(2) **Route utility:** when does avoiding an expensive replacement help, and when does waiting for benefit evidence hurt?
(3) **Execution consistency:** do modeled coordination, route epochs and final acknowledgments preserve the declared logical delivery contract?

The new evaluation freezes a 580-run main matrix and a separate 60-run weak-loss confirmation. Four selected detection cohorts each achieve 60/60 timely trials; the worst per-condition p95 is 43.12 ms. The separate loss cohort corrects reused derived RNG streams without changing native rules or pooling old and new detection counts. In matched policy tests, benefit gating avoids harmful slow-helper replacement, saving 51.26 ms on average relative to direct replacement. It also delays recovery in some persistent faults. The evidence supports a bounded simulation result, not a uniformly faster policy.

The appendices retain the earlier UDP observer and pending-only route experiments as historical component evidence. We do not combine their timings or probe costs with the new RDMA loop. The reconstructed OptCC schedule is a separate comparator; no completed matched test establishes full LIMER superiority over OptCC. This study covers simulation and CPU emulation; physical deployment is outside its scope.

2 Problem Setting and Comparison Scope

2.1 Active-Active Communication and Impairment

A plane is an independently routed network connecting the participating workers. Both planes carry communication slots in the initial plan, so the alternate plane is already active. A slot identifies a sender and a communication step. Local rerouting changes affected slots while retaining the workers and communication schedule. A

*Author list and affiliations to be finalized.

ring schedule has $2(p-1)$ reduce-scatter/all-gather communication steps for p ranks, with endpoint dependencies that let a slow transfer delay subsequent work.

A hard fault removes carrier; a gray fault preserves carrier while impairing delivery. We distinguish random packet loss, configured service-rate reduction and severe delivery delay. Their effects on transport are different. The target decision concerns impaired delivery, including severe slowdown; it need not establish permanent loss or its physical cause. The detectors identify original sender/plane paths, not arbitrary physical hops inside a multi-hop fabric.

2.2 Timing Target and Evidence Boundaries

Detection latency runs from fault onset to the delivered-report-driven decision. In packet simulation, onset is the simulated injection timestamp; in emulation, it is completion of the injection action. Each campaign registers eligible severity and duration and retains every attempt. The primary empirical gate requires at least 95% timely, correctly localized decisions in each eligible condition and no counted control or off-target alarms. This finite-sample gate is not a population confidence guarantee. Earlier all-trial and loss-only gates keep their original verdicts. The new RDMA score requires every affected direction to be timely, with no pre-fault or unaffected-path alarm throughout the registered observation horizon.

Detection, route commit and completed communication are different milestones. We report their respective timings without adding measurements from separate experiments into an end-to-end claim. Hard-fault detection below 1 ms and progress resumed within 1 s of detection remain additional design targets, not demonstrated global guarantees. Numerical AllReduce correctness is also unvalidated: the workloads execute communication dependencies and framed payloads, without performing tensor reduction.

2.3 Related Work and Comparator Roles

OptCC schedules AllReduce under post-failover bandwidth asymmetry [2]. It selects a collective schedule, whereas our route policy selects placement of affected work across active planes. These decisions can be composed. We reconstruct OptCC as an executable scheduling comparator and compare it with an ordinary ring; we do not equate OptCC with staying on a damaged path. The crossed ring/OptCC and route-policy comparison remains unrun.

Chameleon selects data rerouting or dynamic parallelism after node failures [8], so adaptive recovery selection alone is established prior work. SprayCheck uses per-spine distributions under packet spraying to detect gray failures [5]. Its evaluation focuses on spine faults and sketches an access-link count/NACK extension. It is a mechanism reference for our slot-based access observer, not a reproduced head-to-head detector. We do not convert iteration-based detection into a claimed 100-ms comparison.

VCCL uses primary and backup queue pairs for port-failure recovery within its collective library [7]. Varuna distinguishes unexecuted operations from operations whose acknowledgments were lost, using completion logs, extended atomic status and dynamic connected queue pairs [6]. Its configured backup order is different from our remaining-work cost heuristic. Its RDMA microbenchmark and Motor TPC-C results are not training measurements in this study, and its logging corner case and separate CAS argument do not prove TCP replay correctness. These systems motivate explicit execution identity. They are literature comparisons, not reproduced recovery baselines here.

3 Fast Detection from RDMA Delivery Observations

3.1 Inputs and Healthy Calibration

The native detector, `canary-service-v4`, monitors 32 original paths: 16 senders, each with an A and a B path to its ring successor. It receives actual QP submission/completion events, sender retransmissions, business ACK prefixes and receiver-local accepted-packet times. Fault onset, affected node, injected service fraction, drop probability and future recovery time are available only to the injector and offline scorer.

Each path sends a real 64-KiB RDMA canary every 2 ms, starting at 1 ms and continuing to the observation horizon. The first 20 ms freezes the maximum completed canary service time b; this interval is assumed healthy. The deadline is $\max(100\ \mu s, 2b + 100\ \mu s)$, with a 2-ms fallback when no completed calibration sample exists. The fallback does not establish coverage of arbitrary startup degradation.

3.2 Three Complementary Evidence Lanes

Repeated canary impairment. A canary becomes bad when it exceeds its deadline or actually retransmits. An outstanding canary can count before completion. Repeated retransmissions of the same canary count once. At least three distinct bad canaries in a trailing 40-ms window, with submission times spanning at least 4 ms, trigger an alarm. The 2-ms cadence therefore supplies repeated evidence well before 100 ms when the impairment affects canary service.

Business progress stall. An allocated, unfinished original-path chunk with no real ACK-prefix progress for 40 ms supplies a second alarm lane. Idle work, unallocated work and already migrated epochs do not supply this signal. This lane can expose delivery stalls that surviving canaries fail to reveal.

Receiver spacing shift. Completion-time thresholds alone missed capacity reductions in the first formal validation. v4 adds the P10 of positive interarrival gaps between adjacent accepted packets on the same original path, aggregating business and canary QPs. Both packets must have a full 976-byte payload; a short tail breaks adjacency. The 1–21-ms interval calibrates the P10. Thereafter, two consecutive nonoverlapping 10-ms windows, each with at least 64 gaps, must exceed 2× the baseline P10 plus 100 ns. Insufficient samples or a healthy window interrupt confirmation. This is a model-specific service-change rule, not a general capacity estimator or a way to distinguish all congestion from faults.

3.3 Alarm Delivery and Measurement Assumptions

The first qualifying lane emits one alarm per path. A 96-byte alarm pays serialization on a shared 100-Mb/s management FIFO plus

1-ms propagation. Latency is scored at actual controller receipt, including queueing, rather than at the earlier detector callback. Fast evidence accumulation, not post hoc subtraction of control time, explains the measured sub-100-ms result.

Endpoint timestamps and local summaries are ideal simulator observations. The implementation feeds endpoint evidence into a shared detector object; it does not separately serialize every raw observation or spacing summary between hosts. Clock error, timestamp noise and callback CPU time are not modeled. Thus charging alarm transport is not evidence that a distributed telemetry implementation has been validated. The older explicitly reported UDP observer has a different observation contract and remains in Appendix A.

4 Cost-Aware Replacement and Coordinated Execution

4.1 Measure the Candidate Before Choosing It

The candidate route uses three store-and-forward legs: source GPU to a local helper, helper to remote helper over healthy B accesses, and remote helper to the original destination. Candidate eligibility uses modeled phase-idle state and resource credits. It is not an implemented distributed discovery protocol. A full 64-KiB probe measures all three legs; its 96-byte health report must actually arrive through the control FIFO before use. Freshness is measured from physical probe completion, including report transit, with a 20-ms limit. Both compared replacement policies pay the same candidate measurement and health-report costs.

4.2 Compare Finite Remaining Work

For each sender/phase, the controller reconstructs A-owned and B-owned unacknowledged bytes plus flexible pending bytes. Direct service is the ACK progress observed over the previous 10 ms. If a lane has no owned bytes and no recent business progress, a fresh completed canary can supply a service estimate. Observed zero service is not knowledge that the fault will persist.

Let $F(a, b, p; r_A, r_B, d)$ be the fluid finish estimate: A serves immediately, B serves after delay d, owned bytes a, b remain assigned to their lanes, and pending bytes p can use spare service on either lane. The implementation finds the earliest feasible completion time by bisection. With probe leg times ℓ_1, ℓ_2, ℓ_3, final-notification estimate q and chunk size $C = 64$ KiB, the candidate service estimate is

$$\widehat{r}_H = \min\left\{\frac{C}{\max_j \ell_j}, \frac{4C}{\ell_1 + \ell_2 + \ell_3 + q}\right\}.$$

The four-credit limit bounds overlap; it does not divide the first chunk's end-to-end latency by four. The switch estimate charges pipeline fill, current management queueing, four-party coordination and final notification. Even a short rescued tail pays a complete first route/ACK cycle.

The policy compares F under observed direct service with F under the measured helper service and its setup delay. It switches only when estimated gain exceeds $\max(1\text{ ms}, 0.1\widehat{T}_{\text{stay}})$. A finite helper estimate can justify replacing an observed zero-service owned lane. The gate is checked after the paid probe and again before initial commit; rejected attempts wait 5 ms before retrying. This is a sender/phase heuristic, not a global DAG optimizer. Access to modeled helper eligibility and current control-queue state is an explicit simulation assumption.

4.3 Logical Ownership and Completion

PREPARE, READY and COMMIT coordinate the source, destination and two helpers. Messages pay the same 96-byte, 100-Mb/s, 1-ms-propagation FIFO costs as alarms and health reports. Phase and binding version reject stale reports; helper idleness and freshness are rechecked before activation. Replacement keeps the physical A/B topology, retires the old B allocation and rescues the unacknowledged suffix under a new chunk epoch. Old-epoch ACKs cannot advance that logical suffix. Three completed legs and a final source notification are required before logical completion.

The audit checks physical NIC selection, helper and source credits, ACK prefix monotonicity, route epochs and final delivery accounting. These are logical communication invariants, not tensor reduction correctness. A proxy activation or rescued suffix also does not imply sustained use of B′ for new chunks; that outcome is measured separately.

5 Evaluation

5.1 Registered Model, Metrics and Internal Controls

The new packet experiments use the ns-3/RDMA backend in the collaborator's standalone simulator, run on CPUs without real GPUs. The topology has 16 logical GPUs, four per server, physical A/B planes and 10-Gb/s accesses. Normal local helper links are modeled at 2400 Gb/s; the 1-Gb/s helper is an artificial stress boundary, not a hardware configuration claim.

The ordinary workload has two ring rounds, 60 phases and 64 chunks of 64 KiB per rank/phase, totaling 4,026,531,840 logical bytes. Work starts at 1 ms and observation ends at 400 ms. The slow-helper plus quarter-capacity boundary deliberately uses one round, 30 phases and 256 chunks per rank/phase, totaling 8,053,063,680 bytes, with an 800-ms horizon. All arms within a pair share workload and horizon; cross-row speedup comparisons between these workloads are invalid.

We design the fault injections; they are not drawn from a datacenter trace. A chosen GPU's B access receives a blackhole, service fraction 0.25, independent per-packet loss 0.005, or 80-ms channel propagation delay. Placement is drawn from 16 GPUs and onset from 35–55 ms. Both directions of the access are impaired. Extra boundaries use a 2-ms blackhole, 1-Gb/s helper links, or B blackholes on all four GPUs in server 0. The last case removes idle local helper candidates.

The main matrix contains 580 registered attempts: four benefit-policy fault cohorts of 60, 60 healthy controls, 20 three-arm pairs per main fault type and 10 three-arm pairs per extra boundary. The main cohorts already include the benefit arm of each pair. A separate 60-run weak-loss confirmation brings the new total to 640. Native source, binary, parameters and inputs are frozen before execution; development cases are not acceptance samples.

The three internal arms share adaptive A/B allocation and detection: *no proxy* (G10) never replaces B, *direct replacement* (G11)

Table 1: New RDMA-loop detection. Every row has 60 registered trials; all affected directions must be correct and timely. Delay starts at injection and ends at controller receipt. No row has late, missed or invalid trials.

Condition	< 100 ms	p95 (ms)	Max (ms)
Persistent blackhole	60/60	8.64	8.81
B capacity 25%	60/60	29.82	30.02
0.5% loss, new streams	60/60	43.12	47.63
80-ms link delay	60/60	8.66	8.72

coordinates a qualified candidate after alarm and probe, and *benefit gate* (G11) adds the cost comparison. "Direct" does not mean zero-cost or immediate migration. These are mechanism controls, not external-system baselines. Ring versus reconstructed OptCC remains a separate scheduling comparison (Section 5.5); VCCL, SprayCheck and Chameleon remain literature comparisons.

5.2 Detection Meets the Registered Empirical Target

A timely success requires every affected original direction to reach the controller strictly below 100 ms, with no pre-fault or unaffected-path alarm in the full horizon. A usual access fault affects two monitored directions; the no-helper boundary affects eight. Late, missed, invalid and missing attempts remain in each registered denominator. Trial latency is the slowest affected direction, and p95 uses nearest rank over the full denominator, assigning a missing latency infinity.

Each selected condition passes the empirical 95% target (Table 1). The worst observed latency is 47.63 ms; this is a sample maximum, not a guaranteed bound. Per-condition 60/60 gives a one-sided exact 95% lower success bound of 95.13% only under the stated sampling and independence assumptions; it is not a simultaneous multi-condition or arbitrary-fault guarantee.

The loss row deliberately uses the independent confirmation. The inherited injector derives streams from `seed+node`; the main 60 loss trials contain only 38 distinct stream pairs. We retain all original counts but do not use their nominal binomial bound as independent-sample evidence. A separately registered stride-32 cohort supplies 60 non-reused pairs, with unchanged native binary, rules, workload and audit modules. It was registered before the first main weak-loss result. The final row replaces, rather than pools with, the old loss cohort. Policy pairs remain the main matrix's empirical matched comparisons.

All four extra benefit-policy detection boundaries achieve 10/10 timely trials. Healthy controls have 0/60 alarms, but repeat the same deterministic healthy trajectory, not 60 traffic classes. Healthy canary data-side wire load averages 2.83% of the total capacity of the 32 sender-NIC/plane exits, including data headers but excluding transport ACKs; management messages are charged separately. This is not the utilization of an arbitrary fabric link or a complete wire-overhead total.

5.3 When Does Benefit Gating Help?

Policy comparisons use 20 matched triples per main condition and 10 per boundary: identical seed, placement, onset, workload and observation horizon. Table 2 retains all 120 triples. Completion time starts at business launch (1 ms); an unfinished job receives the horizon-based cost of 399 ms, or 799 ms in the longer slow-helper/capacity boundary. The capped mean therefore includes censored observations and must be read with the completion counts, not as an exact mean completion time or speedup.

The slow-helper/capacity boundary isolates the value of refusing a healthy but costly candidate. The gate improves over direct replacement in 8 of 10 pairs and ties in two, reducing the mean by 51.26 ms (11.63%). It is still 0.39 ms slower than no proxy, so the finding is avoidance of harmful replacement, not improvement over the best staying control. Persistent blackhole and weak-loss groups instead pay 5.75 and 11.60 ms additional latency. A gate based on past finite windows can wait for service evidence; it does not know how long a fault will last. There is no uniform policy win.

Without idle helpers all three arms remain unfinished despite timely detection. The short transient also leaves stalled business work in this transport configuration; the replacement arms complete, but the gate does not improve on direct replacement. These are retained boundary outcomes, not cases removed from the performance denominator.

5.4 Execution Audits and Retained Validation Failures

All 640 new runs pass the declared execution audit and independent detection and policy replay. Raw packet events reconstruct receiver P10/counts and canary retransmissions; ACK and physical-leg events reconstruct remaining work, route measurements, health-report delivery and policy arithmetic. All 640 retained raw archives have verified SHA-256 digests, and frozen scoring regenerates the retained trial and pair tables. These checks verify recorded content and modeled invariants, not every job's completion.

In the main four benefit-policy cohorts (240 trials), 180 trials activate a proxy and 1107 chunks begin in-flight rescue. No newly assigned B′ chunk completes, and no trial establishes sustained new B′ use across phases. The current result therefore demonstrates rescue of stalled work followed by adaptive service, not restoration of healthy dual-plane throughput. A no-proxy timeout remains a valid execution outcome.

The first formal `canary-service-v3` validation is preserved separately: 580 attempts, 530 with valid execution/replay and 50 infrastructure errors caused by disk quota. In the quarter-capacity cohort, all 60 executions are valid but only 42/60 detect every affected direction below 100 ms; 18 miss a direction. Counting only the first alarm would conceal these misses. Healthy queueing inflated the canary completion reference. v4 adds whole-path receiver spacing after development showed that an isolated canary train was confounded by QP interleaving. Four old misses are development regressions only; the new matrix uses disjoint input seeds and a fresh frozen build. The first validation is never relabeled as a pass or pooled into the new 640.

Table 2: Internal policy comparison in the new RDMA loop. N/D/G denote no proxy, direct replacement and benefit gate. Every registered triple is valid; completion counts can still be zero. Lower capped cost is better. Negative G−D favors the gate. * marks the longer 799-ms-cap workload; other rows use a 399-ms cap.

	Matched	Completed	Mean capped cost (ms)			G−D
Condition	triples	N / D / G	N	D	G	(ms)
Persistent blackhole	20	0 / 20 / 20	399.00	211.41	217.16	+5.75
B capacity 25%	20	20 / 20 / 20	196.41	196.41	196.41	0.00
0.5% packet loss	20	0 / 20 / 20	399.00	222.76	234.36	+11.60
80-ms link delay	20	0 / 20 / 20	399.00	211.09	216.47	+5.38
No idle helper	10	0 / 0 / 0	399.00	399.00	399.00	0.00
Blackhole, 1-Gb/s helper	10	0 / 10 / 10	399.00	215.53	217.84	+2.31
B capacity 25%, 1-Gb/s helper*	10	10 / 10 / 10	389.15	440.80	389.54	-51.26
2-ms transient	10	0 / 10 / 10	399.00	243.63	243.63	0.00

Table 3: Independent network-only replay, $L = 2$, $k = 4$. Ratio is ring makespan divided by OptCC makespan. These are reconstructed measurements, not reproduced paper figures or LIMER results.

p	N (MiB)	Ring (ms)	OptCC (ms)	Ratio
5	5	0.712	0.638	1.116
5	20	2.754	2.283	1.206
16	15	2.500	1.986	1.259
16	60	9.718	6.917	1.405

The historical standalone UDP observer achieves 400/400 timely faults under its own representative-probe envelope, but its 75.4-ms worst condition p95, probe costs and eight-path traffic model are different. Its earlier low-budget misses, reordering false alarms and management-delay failures remain in Appendix A. Those results motivate the new design; they are neither its baselines nor additional trials in its denominator.

5.5 Executable Scheduling Comparator

OptCC and ordinary ring schedules share the same network runner using SimAI and ns-3. The matched setting has one nonblocking switch and one participating GPU per server. Healthy NICs supply 200 Gb/s; the last-rank NIC supplies 100 Gb/s ($L = 2$). Both arms use 500-ns propagation, 9000-byte payload MTU, DCQCN and 3-μs launch delay. The setting includes no packet loss, NVLink or PXN. OptCC uses $k = 4$, and both schedules transmit $2(p-1)N$ data bytes in aggregate; OptCC synchronization nodes additionally carry 20-byte control payloads.

All 24 primary launches pass; three deterministic launches per arm and condition reproduce the same values. OptCC reduces network makespan by 10.4–28.8% relative to ring in these four settings. Four healthy-ring references and two segmentation checks bring the measurement denominator to 30/30. These repeated fixed settings do not provide confidence intervals.

The public artifact lacks the original simulator integration and experiment driver, so this is an independent reconstruction, not an exact reproduction of a published performance figure. It omits native collective memory and reduction costs. Appendix D preserves artifact, transport, scheduler and segmentation checks. The result establishes an executable schedule comparator, not LIMER superiority. The planned common-backend matrix crosses ring/OptCC schedules with stay/fixed-switch/LIMER route policies under identical faults, workload and seeds, charging all recovery cost and retaining failures. That joint comparison remains unrun. The new 16-GPU helper-recovery backend and the earlier reconstructed scheduling runner are not a matched full-system comparison.

6 Limitations and Remaining Validation

The detector and helper recovery are now evaluated together in a native packet-level simulation. This closes the earlier separation between isolated detection and route components, but only under the newly registered model. It does not establish general fast gray-fault coverage. The pass depends on a healthy 20-ms startup reference, fixed canary geometry, ideal nanosecond endpoint timestamps and centrally available local summaries. Raw observation exchange, timestamp noise and detector processing time are not separately charged. Only alarms and the declared recovery/control traffic pay the management FIFO. Arbitrary background traffic, startup degradation, correlated faults and multiple episodes remain unvalidated.

The healthy control repeats one deterministic trajectory. The selected weak-loss detection cohort fixes the derived-stream reuse issue, while main policy pairs remain empirical comparisons under the original stream schedule. The confidence bound is per condition under sampling assumptions, not a guarantee over all fault distributions. Earlier failed validations remain failures.

Benefit gating approximates sender/phase completion from recent service and candidate probes. It assumes access to modeled helper eligibility and control-queue state; distributed discovery and uncertainty-aware congestion prediction are not implemented. The policy loses time under persistent blackhole and weak loss, and the favorable slow-helper case uses a deliberately extreme local-link boundary. No idle helper means detection cannot produce recovery. Sustained new B′ throughput, candidate secondary failure and restoration of old B have not been demonstrated.

Execution checks establish logical epochs, prefixes and final-delivery accounting. The simulator does not execute tensor reduction, CUDA, HBM or compute/communication overlap. There is no completed full LIMER versus OptCC/VCCL comparison or 128-rank

validation. The earlier UDP and pending-only experiments retain separate time bases and are not combined with the new loop into additional end-to-end evidence. Real-GPU execution and deployment are outside the agreed simulation scope.

7 Conclusion

LIMER now implements a connected RDMA simulation loop for timely impairment detection, measured helper-route benefit decisions and coordinated rescue. Four selected registered detection cohorts each achieve 60/60 correctly localized controller alarms below 100 ms, with a worst per-condition p95 of 43.12 ms. Matched policy experiments show both the benefit of avoiding an expensive helper route and the latency cost of waiting under persistent faults. Full denominators, independent replay, the failed first validation and explicit measurement assumptions bound these findings. General gray-fault coverage, sustained replacement-plane throughput and external full-system comparisons remain open simulation work.

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A Historical Standalone Observations and Route Components

This appendix preserves the earlier methods, results and original verdicts. Here "primary detector" refers to the historical UDP observer, not the new RDMA detector in Section 3. The pending-only route loop also has a different execution contract from the new epoch-based rescue. These campaigns are not pooled with the 640 new runs.

A.1 Historical UDP Observation Design

A.1.1 Observation Contract. The primary candidate detects delivery impairment from observations available at the two ends of each monitored path. Business packets and probes have separate accepted/delivered counters and independent source sequence numbers. Reports carry path, batch, traffic class, sample time, arrival time and receiver gap evidence. Injection time, affected-path truth, private drop counters and future traffic never enter the decision interface. The collector runs the rule on report arrival, so reporting delay contributes to the measured confirmation time.

Business and probe classes remain separate because successful probes can otherwise hide business impairment. The receiver starts a sequence gap only after a higher sequence arrives and clears it when the missing packet is later delivered. A gap therefore records incomplete delivery rather than proving permanent packet loss. The observation identifies a path in the declared one-hop model.

A.1.2 Guarded Impairment Decision. A receiver gap persisting for at least 40 ms supplies impairment evidence. Its age is computed at sampling, without counting report transit as extra receiver evidence. Two aligned business-only service windows also detect a sustained delivery decline that healthy probes could mask. The guarded service rule requires sufficient accepted traffic and a delivered increment no greater than 60% of the accepted increment in two consecutive windows. The aligned early lane uses at least 32 accepted packets; missing, invalid or nonsevere windows break evidence.

The candidate retains the frozen hybrid observer's mature batch-deficit and aligned total-service lanes as fallbacks. Their exact guard, window and reset rules are described in Appendix B. Only the first alarm per path is emitted in a lifecycle. The frozen hybrid rule is also scored independently on the same delivered reports as a shadow baseline. This shared input permits rule attribution within a campaign; results from different traffic models are not treated as a matched ranking.

A.1.3 Representative Probes and Causal Compensation. Representative probes use the business packets' 1500-byte wire size and the same modeled socket, queue, route and application delivery stage. This common fault treatment is an explicit assumption. Same-path small probes do not imply equivalent impairment exposure: the matched selective-loss contrast in Section A.3 demonstrates the distinction.

Sparse or idle business traffic provides too few observations, so a source policy adds probes according to accepted traffic in the previous closed 10-ms business window. Let D_{b-1} be that count and Δ_b the next source span. The source evenly schedules

$$P_b = \max\left(0, \left\lfloor \frac{25000\Delta_b}{1\text{s}} \right\rfloor - \left\lfloor \frac{D_{b-1}\Delta_b}{10\text{ms}} \right\rfloor\right)$$

probes in that span. Startup from 1 to 10 ms assumes no prior business. This policy uses neither receiver loss nor fault metadata. Historical traffic can cause transient overshoot or lag, and every actual emitted probe byte is charged. A constant 35,000-pps representative source is evaluated separately to expose the effect of observation budget.

Table 4: Causal traffic-compensation validation on 10-Gb/s links. The detector and 40-ms guard are unchanged; each attempt stays in its denominator.

Condition	< 100 ms	Late	Miss	P95 ms
Selective, 80 Mb/s	60/60	0	0	70.1
Application loss, 80 Mb/s	60/60	0	0	72.1
Selective, 1 Mb/s	60/60	0	0	73.6
Business idle	60/60	0	0	75.4
Selective, 1 Gb/s	60/60	0	0	58.7
Application loss, 1 Gb/s	60/60	0	0	57.5
Network delay 80 ms	20/20	0	0	54.5
Application delay 80 ms	20/20	0	0	54.3

A.1.4 Reporting Cost and Output. Source and receiver clocks are aligned in the evaluated model. Reports occupy 96 bytes per copy: 68 payload bytes plus modeled IPv4/UDP overhead, without link-layer overhead on the abstract collector. A separate serialized 10-Mb/s collector adds 1-ms propagation. All 4800 offered report copies per lifecycle cost 460.8 KB, with 7.3728 Mb/s steady offered load. This collector does not share congestion with the monitored data paths. Its capacity and delay are part of the observation envelope. Callback computation time is not modeled as elapsed simulation time.

The detector outputs an impaired path and decision time for subsequent route-cost evaluation. No route action follows from the detection experiment itself. Multiple-episode rearming and arbitrary report loss, management delay or unknown probe treatment remain outside the latest validation.

A.2 Fast Detection in the Registered Envelope

The primary validation contains eight independent one-hop UDP paths with 10-Gb/s links, a fixed affected path, 250-ms source activity and observation drain through 600 ms. Six weak-fault conditions use 0.5% persistent loss: selective network/application loss at 80 Mb/s and 1 Gb/s business, sparse 1-Mb/s selective loss, and idle size-independent loss. Two severe-delay conditions add 80 ms to 25% of network or application deliveries. All paths are monitored; fault time and identity enter scoring only.

After 130/130 timely development faults and 30 alarm-free development controls, unchanged-source fresh validation produces 400/400 timely fault decisions (Table 4). Every condition meets the empirical target, with no late or missed decisions and largest per-condition p95 75.4 ms. The 60 primary controls and a separately registered 60-case high-traffic supplement have no alarms. Controls include source pauses and 40-ms network/application deferrals; these finite checks do not establish a population false-alarm rate.

Independent replay reconstructs counters, gap ages, probe emissions, encoded reports, collector service and native/Python alarm agreement. The result is conditional on aligned clocks, representative fault treatment and the stated reporting capacity. Even 60/60 and 20/20 successes have two-sided Wilson 95% lower bounds of only 93.98% and 83.89%, respectively. An empirical pass therefore does not establish a population 95% guarantee or a collective recovery benefit.

Table 5: Actual average adaptive probe traffic on each 10-Gb/s path. The denominator includes the full source lifecycle, not only steady state.

Business profile	Probes (Mb/s)	Link capacity (%)
Continuous 1 Gb/s	10.84	0.108
Continuous 80 Mb/s	223.18	2.232
Sparse 1 Mb/s	299.04	2.990
Business idle scene	274.60	2.746

A.3 Observation Cost, Attribution and Retained Failures

Causal traffic compensation reduces extra observations when business traffic already supplies them. Table 5 charges actual accepted probe bytes over the full 249-ms source interval, including startup and lag. At 1-Gb/s continuous business, only 225 startup probes/path are emitted; sparse and idle business still require substantial active sensing. The separate 7.3728-Mb/s reporting load is additional to these probe costs.

Same-report comparisons distinguish observation policy from decision-rule effects. Under constant 1500-byte probes at 35,000 pps, nominal probe cost is 420 Mb/s/path, or 4.2% of link capacity. Candidate validation gives 400/400 timely faults, while the frozen hybrid shadow gives 396/400 and also meets every condition's 95% gate. Most of that target pass therefore reflects the changed observation opportunity. Under adaptive observations, candidate validation again gives 400/400 while its same-report shadow gives 391/400; idle coverage is 60/60 versus 55/60. The guarded gap rule supplies a target-relevant difference in that adaptive campaign. Constant and adaptive campaigns use fresh ledgers; they are not paired-seed algorithm-only arms.

Probe representativeness and budget remain necessary conditions. An earlier matched 80-Mb/s contrast has the same business send/loss counts and 123 dropped business packets in total in each arm: the small-probe hybrid detects 20/20 IID-loss cases but only 5/20 size-selective cases while its probes stay healthy. At the original 100-Mb/s, 17.92-Mb/s probe envelope, class-separated development retains 16/20 timely selective and 13/20 application-loss decisions. Making probes business-shaped at the old byte budget reaches 20/20 selective decisions but only 18/20 application-loss and 2/5 idle decisions. These low-budget conditions retain their original FAIL verdicts. Appendix C retains all earlier detection tables, guard sensitivity, controls and setup failures.

A.4 Historical Pending-Only Route Design

The following components study how an impairment decision can be used. They are evaluated separately from the primary detector. The bounded online route loop uses native sender ACK progress, not the representative packet detector in Section A.1. This distinction is retained in every performance interpretation.

A.4.1 Remaining-Work Switch-versus-Stay Heuristic. Let I_A, I_B be launched, unacknowledged bytes and W_A, W_B pending bytes on the two planes. Only pending affected A work can change routes. The current observed ACK service is r_A; alternate service r_B is the minimum of its calibrated service and its last positive fully busy

sample. Unknown, zero or more-than-20-ms-old alternate service selects stay. Otherwise the sender-service heuristic compares

$$\widehat{T}_{\text{stay}} = \max\{(I_A + W_A)/r_A, (I_B + W_B)/r_B\},$$

$$\widehat{T}_{\text{switch}} = \max\{I_A/r_A, (I_B + W_B + W_A)/r_B\} + c.$$

The registered cost budget is $c = 10$ μs, exceeding the default modeled transaction cost. A switch requires pending A work and a smaller switch estimate. Remaining work on B is charged because both planes are active. The estimate is a sender-service proxy, not a collective critical-path optimizer or a prediction of future alternate congestion.

For this component experiment, native sender ACK credit combines completed flow bytes and active acknowledged bytes. Samples are taken every 2 ms and reports pass through the modeled collection channel. A detector calibrates four positive fully busy samples and alarms after three busy samples below half their reference; idle windows interrupt evidence. These configured-rate development checks do not establish the primary detector's broader weak-loss/idle coverage. Appendix F retains the full detector and earlier policy details.

A.4.2 Versioned Activation and Transfer Ownership. At transfer launch, the SimAI target selects the committed plane, posts one matching receive and retains that transfer's ports and version through completion. A route update changes only pending affected transfers. PREPARE, READY and COMMIT messages coordinate a common version and activation point across all four ranks. The reliable modeled management FIFO charges 64 bytes per message with 1-Gb/s serialization and 1-μs propagation by default. Its messages are not packetized on the RDMA data network. A missing READY prevents activation; a nonconsecutive version is rejected.

Launched transfers retain ownership on their original paths. Consequently, the pending-only mechanism cannot rescue a severely stalled launched transfer. The separate CPU/TCP harness checks frame round, step, route version, payload length and checksum, with explicit ownership of partially read frames. These transport checks are documented in Appendix E; they are not an exactly-once numerical reduction proof and do not imply transparent migration of in-flight work.

A.5 When Localized Switching Helps

These experiments evaluate route utility separately from the primary detector. A static four-rank, 3-MiB ring calibration fixes both access rates across stay/switch arms and changes only the affected rank's A sending slots. At 40/200 Gb/s, switching reduces completion from 0.740 to 0.571 ms (22.9%); at 40/20 Gb/s, it increases completion from 1.454 to 1.934 ms (33.0%). Table 6 excludes runtime detection, coordination and replay costs. It motivates the route decision but is not a controller result.

The bounded online comparison uses the ACK-progress detector and remaining-work heuristic of Section A.4. Four ranks execute fourteen 3-MiB rings with prescribed releases at 0,100,...,1300 ms. At 312 ms, rank-2 A configured egress service falls from 1 to 0.2 Gb/s; B initially supplies either 1 or 0.1 Gb/s. Other accesses remain at 1 Gb/s. All four arms share workload and per-report cost, with 96-byte path reports offering 3.072 Mb/s to the 10-Mb/s collector. Each activated plan charges 768 modeled control bytes.

Table 6: Static dual-plane network calibration. Both route arms share each row's capacities; control, cutover and replay costs are excluded.

C_A (Gb/s)	C_B (Gb/s)	Stay (ms)	Switch (ms)
100	200	0.352	0.305
40	200	0.740	0.571
40	20	1.454	1.934

Table 7: Bounded online ring comparison: total network-DAG completion from workload start (ms). Fixed requests localized switching at startup; oracle supplies fault timing/identity, not optimal route choice. All arms share workload, reporting model and charged coordination.

C_B (Gb/s)	Stay	Fixed	Oracle	Candidate
1.0	1545.674	1339.485	1339.485	1339.485
0.1	3958.544	5360.714	5190.811	3958.544

All eight registered workloads complete. With faster B, the candidate alarms after 9.2304 ms, changes 31 pending transfers and completes in 1339.485 ms rather than stay's 1545.674 ms, a 13.34% reduction. With slower B, it alarms after 35.2304 ms and stays, completing in 3958.544 ms; fixed switching instead takes 5360.714 ms, a 35.42% increase over stay. Neither candidate case has a pre-event or unaffected-path alarm. Independent replay audits 96,224 delivered reports, 2.114 GB of ACK credit, policy arithmetic, dependencies and immutable launched ownership.

These deterministic configured-rate examples establish bounded route selection behavior. They do not establish weak-loss coverage, optimality, future alternate-capacity guarantees or performance with the primary representative detector. Appendix G retains the static setup corrections and severe-loss TCP contrast.

A.6 Execution Consistency Checks

The scripted SimAI transaction gate has nine prospectively registered engineering cases: eight workloads complete with transport-map drain and one deliberately nonconsecutive version is rejected. The default request at 5 μs activates at 13.145 μs, moving two still-pending affected transfers while launched work retains its endpoints and version. Missing READY prevents activation. With slower coordination, the workload can finish before activation, so a valid update moves no transfers.

Independent reconstruction checks all 582 trace records and the frozen dependencies. Separate framed-TCP runs check integrity and version agreement, but retain transient false commits and severe-loss timeouts. Appendix E gives those failures and the complete transaction cases. These checks validate the declared execution contract, not numerical reduction, cancellation/replay or rescue of already-launched stalls.

B Alternative Observation Designs

This appendix records earlier observation components used as references or fallbacks. They are not interchangeable versions of one

evaluated joint system. Their original traffic models and failed gates remain unchanged.

B.1 Counter, Transport and Hybrid References

Hard faults use switch-facing link-state notifications. For gray faults, the prototype samples switch-facing receive counters at a nominal 20-ms interval, calibrates healthy burst rates and flags degraded bursts or deep stalls. A worker step gate then compares affected communication to fault-free per-slot timing. The observer runs in the management process, not a switch ASIC, and injector qdisc state is excluded from its inputs.

An opt-in Linux per-socket retransmission sampler records cumulative counters, deltas, connection identity and missing reads on the same time axis. It is observation-only. Positive retransmission deltas can also arise from congestion or reordering, and a delayed first delta cannot establish a 100-ms bound. This passive observation does not validate the separate hybrid detector or connect it to automatic recovery.

We additionally implement an isolated passive host decision using extended Linux `TCP_INFO`. ACK-based recovery and reordering feedback motivate this observation choice [3]; the rule interprets Linux state rather than implementing RACK-TLP. Every outgoing A/B connection is sampled continuously at a nominal 5 ms, without injector timing or location as input. The frozen rule requires at least two retransmitted segments in a trailing 100-ms window, outstanding data and 30 ms since initial loss evidence. A new DSACK or reordering feedback event vetoes a decision for 100 ms. Missing reads, gaps above 15 ms and connection/counter discontinuities clear evidence; repaired holes alone do not erase recent repeated-loss evidence. Each connection generation can issue one decision. This identifies a sender/fabric transport path, rather than a physical hop, and remains disconnected from automatic rerouting.

An isolated packet-simulation candidate adds two-ended batch counters, motivated by alternate-marking measurement and its guard requirements [4]. This is a custom observation model, not a reproduction of that standard. Each 1500-byte wire packet carries a 12-byte batch/sequence header. Source acceptance and destination delivery snapshots are independently reported every 10 ms, with a 50-ms receiver guard and explicitly charged report transport. Only path, batch, side, count, sample time and report arrival time enter the detector. It pairs complete batches and requires two missing deliveries within a trailing 100-ms report-acceptance window. Two active zero-deficit batches clear history, including before confirmation, and rearm the rule; idle batches do not prove recovery. Missing or invalid batches interrupt evidence, and reports older than 200 ms after batch closure are unusable. Decisions include report delay and are produced by chronological input replay. This observes a delivery deficit under the guard, without proving permanent loss or its cause. It remains disconnected from a dynamic route controller.

The hybrid candidate retains this mature lane and adds continuous 64-byte wire probes on the same UDP socket, queue, path and receiver as data. Probes run at 35,000 packets/s/path independently of fault time, including during data pauses. Size-independent IID packet loss affects both traffic types. A separate early lane samples cumulative accepted counts and delivered counts simultaneously at closure plus 10 ms. It compares adjacent complete pairs and alarms after two consecutive windows with at least 32 accepted packets and delivered-count increments no greater than 60% of accepted-count increments. Gaps, invalid or unavailable pairs and nonsevere windows break the evidence. Both cumulative samples are monotonic and share the same timestamp; RX credit cannot exceed simultaneous accepted traffic. This early signal concerns delivery service, not identified permanent loss. The observable guard identifies the lane; fault truth remains scorer-only. The combined rule retains the first alarm per path in each lifecycle; multiple-episode rearming is not validated. Both lanes execute natively on delivered-report events, with independent Python replay.

C Additional Detection Evidence and Development Records

The following records retain prior experiments and negative results. The primary candidate and validation are in Sections 3 and 5.2. Cross-model counts below are not pooled into a detector ranking, the historical UDP 400-fault denominator, or the new RDMA validation.

C.1 Transport Diagnostics and Earlier Validation

An observation-only TCP retransmission pilot runs three independent lifecycles at each configured loss level (0.5, 1, 2, 5, 10 and 25%), three short healthy references, and three repetitions each of matched nominal, paced-B-congestion, round-pause and A-reordering controls. All 33 attempts remain in the ledger: 30 complete, whereas the three 25% loss attempts exceed the 120-s round wait. Across 615,998 socket samples, every counter read is available. Availability does not imply that the signal identifies the cause of retransmission.

For the affected rank-2 A connection, the first positive counter delta arrives 44.5–1158.1 ms after 0.5% loss injection, 59.4–157.4 ms at 1%, 59.9–152.6 ms at 2%, and 47.7–62.7 ms at 5%. These are ranges of three raw signal observations per level, not detector decision latencies or confidence bounds. They already preclude a uniform 100-ms claim for the present observer.

Neither nominal nor round-pause controls produce a positive delta; each congestion control produces at least one delta somewhere in the observed sockets, and each reordering control produces 730–892 retransmissions on rank-2 A. Congestion traffic is paced at 50 Mb/s, with measured sender rates of 49.95–49.97 Mb/s. These controls configure no random packet loss; queue drops are not independently ruled out. The existing rate-counter detector emits at least one suspect in all 12 matched controls, which is not a final host-confirmed recovery alarm. The pilot observer remained disconnected from the controller. Its raw signals motivate the independently validated host candidate below; they are not reclassified retrospectively as successful decisions.

Packet-granularity diagnostics. Eight exploratory capture runs compare original veth settings with TSO/GSO/GRO disabled, once each at 0, 0.5, 1 and 2% configured loss. All complete with zero kernel capture drops and windows covering the first 100 ms after injection. Original settings expose IP units as large as 37,700 bytes; disabling offloads bounds observed IP lengths at 1500 bytes but

Table 8: Independent TCP host decisions under the symmetric ingress recipe. All registered fault attempts remain in the timely denominator. P95 is the nearest-rank delay with nondetections assigned infinity.

Loss (%)	< 100 ms / trials	Misses	P95 (ms)
0.5	14/20	0	408.2
1	19/20	0	85.2
2	19/20	0	60.9
5	20/20	0	49.0

still yields a 1253.7-ms first receive sequence gap in the 0.5% case. Disabling device offloads alone consequently does not resolve the latency constraint or validate the injector's loss unit.

A distinct four-run diagnostic moves loss injection to an ingress IFB after sender segmentation, retaining the shaping and disabling offloads. The healthy run shows no sequence gap or positive retransmission delta; one run each at 0.5, 1 and 2% loss shows first receiver gaps at 10.6, 10.9 and 9.2 ms and first positive socket counter samples at 22.3, 22.4 and 19.7 ms. These are exploratory signal times under a changed fault placement, with no new controller trigger or held-out false-alarm validation. Reordering can also create sequence gaps. The captures cover an eight-second prefix, not the full workload. Packet-field checks agree with an independent native decoder in four selected captures; startup failures, incomplete termination statistics and lossy capture attempts remain separately archived. These diagnostics motivate prospective testing without replacing the original measurements or establishing sub-100-ms detection.

Frozen passive host-rule validation. Development distinguishes emulator corrections from detector changes. The first scene produces six nominal false alarms; a separate recipe adds symmetric 100-Mb/s, 1-ms post-segmentation ingress shaping with limit 10,000 to all eight sender paths, retains switch-to-worker shaping, and disables veth offloads. The changed model is not an algorithm-only gain. A continuous-loss rule also misses promptly repaired low-rate losses, motivating the frozen recent-evidence rule. These exploratory trials and failed setup/reporting attempts remain archived separately from the prospective matrix.

The independent matrix registers 20 lifecycles each at 0.5, 1, 2 and 5% loss plus five each nominal, paced-B-congestion, round-pause and 10-ms A-reordering controls. Four ranks exchange framed 2-MiB chunks over one warmup, two baseline and four post-event rounds, with recovery off. Source, wrapper, container image and configurations are frozen. Kernel loss draws are uncontrolled and unpaired; seed 20261010 shuffles launch order only. All outgoing connections are monitored before and after injection. Offline scoring counts the intended rank-2 A path; the detector receives neither its identity nor injection time.

100 of 100 registered lifecycles complete with framed transport integrity. The actual host decision, measured from injection completion, is below 100 ms in 72 of 80 fault attempts; 8 are late and 0 have no correctly localized post-injection decision. P95 and timely fractions are both retained: a p95 pass does not mean every trial passes. The original all-trial deadline gate is FAIL. There are 0 other

Table 9: Frozen two-ended observation rule in isolated packet simulation. All registered loss attempts remain in the denominator; nondetections remain in nearest-rank p95 as infinity. Rates are offered wire Mb/s.

Rate	Loss (%)	< 100 ms	Late	Miss	P95 (ms)
80	0.5	7/20	12	1	228.2
80	1	12/20	8	0	138.2
80	2	20/20	0	0	88.0
80	5	20/20	0	0	78.6
1000	0.5	20/20	0	0	72.5
1000	1	20/20	0	0	68.0
1000	2	20/20	0	0	69.2
1000	5	20/20	0	0	68.1

or pre-injection alarms in fault trials. The 20 primary controls produce 0 false alarm events over 178.4 s of observed lifecycle time; all pre-event and unaffected-path alarms count, rather than only the known faulted connection.

Five separately registered 40-ms reordering stress lifecycles produce 3 false alarm events in 3 of five runs. These no-random-loss failures remain visible and block a broad causal fault-detection or automatic-recovery claim. A sender/fabric decision cannot localize the physical hop, cover an idle path or establish a GPU/RDMA hardware guarantee. The primary gray-fault constraint remains unresolved despite the bounded timing measurements.

Two-ended packet-simulation validation. Eight independent one-hop UDP streams use four rank labels and two plane labels; they do not execute a collective or a training topology. Low and high profiles offer 80 Mb/s on 100-Mb/s links and 1 Gb/s on 10-Gb/s links, respectively, with 1-ms propagation. Packets are paced from 1 to 250 ms; observations continue through 600 ms. Offered rates include all 1500 wire bytes, including the marking and the pinned backend's 14-byte PPP header. Receiver snapshots wait 50 ms after batch closure. A separate modeled FIFO charges 64 bytes per report, 1 Mb/s service and 5-ms propagation; it is not a measured control network and shares no congestion with the monitored data paths. Shared clocks and complete local counters are explicit assumptions.

After eight development trials, we freeze the rule, parameters, binary and simulator libraries before 200 fresh lifecycles: 20 each at four loss rates in both profiles, plus five each nominal, below-capacity congestion, source-pause and 40-ms reorder controls per profile. Fault or pseudo-event phase is independently registered in 50–60 ms. The seed is 20261010, with separate assigned streams and unique validation runs. All eight paths are observed from startup; fault timing, identity and private drop counters enter only scoring. Nominal repeats are deterministic checks, not independent evidence for a stochastic false-alarm distribution.

All 200 primary lifecycles pass final packet accounting. The rule produces 139 of 160 correctly localized loss decisions below 100 ms; the all-trial deadline gate is FAIL. The 40 primary controls produce 0 false alarm events over 24 simulated lifecycle seconds (192 path-seconds). Fault trials produce 0 pre-event or unaffected-path alarms. These simulated times are not pooled with Linux wall-clock

measurements, and the different UDP workload is not a matched comparison establishing improvement over the TCP rule.

All 25 separately registered stress lifecycles drain with audited packet accounting. Under the original loss-only scoring, 80-ms reordering and 160%-capacity competing load each trigger alarms in five of five no-permanent-loss controls. Sparse 1-Mb/s data yields zero timely decisions and five nondetections at 0.5% loss. Idle traffic yields zero timely, one late and four missed decisions; 10% report loss yields zero timely, two late and three missed decisions, with 361 of 2000 scheduled batch pairs unavailable. An unavailable report, including both reports missing, is never interpreted as zero received data. A packet arriving beyond the guard is indistinguishable from a permanently lost packet at the decision deadline; the original control failures remain visible.

Each lifecycle schedules 800 reports, or 51.2 KB of modeled control traffic. During steady reporting, the eight paths offer 819.2 Kb/s, 81.92% of the collector service rate; marking uses 0.8% of each wire packet. These are modeled byte and service costs. Exact replay recomputes all 179620 archived reports, decisions and scores, checking source, binary, library and input hashes. Final UDP packet conservation does not establish numerical reduction or coordinated cutover correctness. An initial 1512-byte wire setup failure and its seven unrun development registrations are preserved; the corrected eight-run development ledger precedes the fresh validation.

Frozen hybrid delivery detection. The archived failures above motivate a new prospective packet-model campaign. Development v1 retained sparse-loss and overload deadline failures; v2's early age-deficit lane alarmed in a mild-congestion control. V3 replaced that lane with delivery service rate, but its first independent validation false-alarmed on a high-throughput source pause. V4 aligns TX/RX sampling times and rejects offset samples before fresh development and validation. All failed ledgers remain available. A separate stale-binary setup failure retains eight failed and 129 unrun attempts before corrected compilation. The frozen validation has 740 fresh lifecycles: 60 each at four loss rates in both profiles and in three weak-loss stress conditions, 20 each for severe delay and overload, and 40 negative controls. Eligible slowdowns are prospectively defined as 80-ms additional delivery delay on 25% of packets or 80%-capacity additional competing traffic. Loss, delay and overload remain separate conditions; the slowdown labels do not rewrite M8's loss-only failures. Run IDs 41000+ and hidden event phase in 50–60 ms are registered before execution. All eight paths are monitored; the injected path is fixed at 2/A. The detector sees neither injection time nor path truth.

All 740 lifecycles pass packet and report accounting with exact native-versus-Python alarm agreement. The hybrid rule yields 698/700 correctly localized fault decisions below 100 ms; every registered eligible condition meets the 95% empirical target. The largest per-condition p95 is 80.7 ms. The 40 controls produce zero alarms, and fault trials produce zero pre-event or unaffected-path alarms. Deterministic nominal repetitions are implementation checks, not a stochastic false-alarm distribution. All scheduled observations, including wholly missing report pairs, are accounted for. The separate all-trial/no-unavailable-report gate remains FAIL; it is not replaced by the empirical fraction rule. Wilson two-sided 95% intervals accompany the released ledger: 60/60 has a lower bound of 93.98%, and 20/20 has 83.89%. Thus an empirical pass does not prove a population lower bound of 95%.

Table 10: Hybrid packet-model validation. Labels r/p give data wire Mb/s and loss percent. Every fault attempt remains in its condition's denominator; misses remain in p95 as infinity. Sparse, idle and report-loss conditions have 0.5% packet loss.

Condition	< 100 ms	Late	Miss	P95 ms
80/0.5	60/60	0	0	78.5
80/1	60/60	0	0	70.5
80/2	60/60	0	0	64.8
80/5	60/60	0	0	62.8
1000/0.5	60/60	0	0	66.8
1000/1	60/60	0	0	64.0
1000/2	60/60	0	0	63.3
1000/5	60/60	0	0	62.4
Sparse 1 Mb/s	60/60	0	0	79.5
Data idle	59/60	1	0	80.7
Reports 10% loss	59/60	1	0	77.7
Delay 80 ms	20/20	0	0	61.8
Overload	20/20	0	0	34.0

The result incurs substantial observation cost: 17.92 Mb/s of probes per path, or 17.92% of a 100-Mb/s link. With 80-Mb/s data, the nominal total is 97.92% even without competing traffic. The 10%-capacity background control therefore offers mild overload rather than below-capacity traffic; it still produces no severe-impairment alarm. Early and mature samples each receive three independently lost 64-byte copies on a separately serialized 10-Mb/s, 1-ms collector. All 4800 copies are charged, including lost copies: 307.2 KB/lifecycle and 4.9152 Mb/s steady load, 49.152% of collector capacity. The workload stops at 250 ms and observations drain through 600 ms. This collector does not share data-plane congestion. The gain over M8 combines changed observation traffic, report service and rules; it is not a matched algorithm-only ablation.

Fifteen fresh boundary trials preserve failures outside the primary envelope. No-probe idle traffic gives 0/5 timely decisions; 120-ms report propagation gives 0/5, and size-selective loss affecting only packets over 128 bytes gives 0/5. The delayed-report boundary observes until 800 ms to drain its tail. In the sparse selective boundary, 4/5 lifecycles realize no business-packet damage; these configured-fault outcomes are not that many demonstrated failures to detect actual damage. A separate active-load supplement uses 20 matched seeds at 80 Mb/s. IID and selective arms have identical business send/loss counts, 123 dropped business packets in each arm, and 0 zero-damage pairs. IID gives 20/20 timely alarms; selective gives 5/20 while all probes remain healthy. All 40 lifecycles pass accounting and exact native replay. This matched contrast demonstrates an actual probe-representativeness limitation, not an assumption of uniform fault coverage. Same path alone is insufficient. These are one-hop UDP observations with a fixed fault location, not an RDMA collective or joint route-control result.

Representative business-delivery observations. The size-selective failure motivates class-separated counters and receiver gaps, without discarding the earlier results. Corrected 100-Mb/s development

Table 11: Constant representative-probe validation on 10-Gb/s links. Weak faults use 0.5% loss; delays affect 25% of packets. All attempts remain in the denominator, with zero late or missed decisions.

Condition	< 100 ms	P95 ms
Selective, 80 Mb/s	60/60	63.2
Application loss, 80 Mb/s	60/60	61.2
Selective, 1 Mb/s	60/60	64.9
Business idle	60/60	62.6
Selective, 1 Gb/s	60/60	56.1
Application loss, 1 Gb/s	60/60	55.9
Network delay 80 ms	20/20	53.0
Application delay 80 ms	20/20	54.6

has 16/20 timely selective decisions and 13/20 application-loss decisions, with zero alarms in 65 controls. The same-report shadow gives 7/20 and 4/20. One application-loss registration realizes no business drop; it remains a miss in the denominator. Lowering the 40-ms gap guard retrospectively false-alarms in all ten lifecycles of each dense, sparse and application 40-ms delay control. The initial setup also incorrectly injected 80 ms under a sparse-40-ms label: its 19 completed, five interrupted and 125 unrun registrations remain archived and supply no acceptance evidence.

Changing probe shape at the old nominal 17.92-Mb/s byte budget improves selective development to 20/20, but leaves application loss at 18/20 and idle loss at 2/5. Smaller probes on fresh, unpaired draws give 17/20, 14/20 and 5/5, respectively. Both arms retain full-coverage FAIL and all 60 controls remain alarm-free. Actual packet-boundary costs are 557,760 versus 558,000 probe bytes/path; this difference is charged.

We therefore register an explicitly different envelope: 10-Gb/s links, 1500-byte business-shaped probes and the unchanged 40-ms detector guard. Constant 35,000-pps probes cost 420 Mb/s/path (4.2% of link capacity), or 525% and 42% of business bytes at 80 Mb/s and 1 Gb/s. Following 70/70 timely development faults and 30 alarm-free controls, the unchanged source passes fresh 460-case validation with 400/400 timely faults and 60 alarm-free controls (Table 11). The same-report shadow baseline gives 58/60 in each 80-Mb/s selective and application-loss condition, and all other fault decisions timely: it also passes every condition's 95% target. The improvement largely comes from observation opportunity, not an established novel decision rule.

D OptCC Reconstruction and Artifact Checks

Section 5.5 reports the matched schedule timings. The following original artifact and implementation records establish how that independent comparator was constructed.

Artifact validation completed. We archived the six-file public reference artifact and compiled its three standalone C++17 schedule generators [1]. Four example cases completed and produced parseable DAGs encoding flow dependencies (Table 12). The single- and multiple-straggler examples passed the generators' internal validation. The multi-GPU example matched the driver's reference makespan; this is a model-level check, not a measured network completion time. The latter example also assumes NVLink-rich servers and is retained only as an artifact check.

Table 12: Completed OptCC artifact checks. Flow counts are generated DAG entries, not GPU execution results or reproduced performance measurements.

Reference case	Flows	Check
Single straggler, $L = 2$	132	PASS
Single straggler, $L = 1.5$	157	PASS
Two stragglers, $L = (2, 1.5)$	274	PASS
Four GPUs/server, $L = 2$	198	Ref. match

Independent packet-level reconstruction. The simulator integration and experiment driver are absent from the public snapshot, preventing an exact reproduction of a published performance figure. We pinned SimAI commit `f5efb5a` and its ns-3 backend `7e3cb5b`. The original simulator's eight-GPU NVLink microbenchmark completed before the graph importer was added. For the single-GPU-per-server setting, our standalone replay uses SimAI's network frontend and RDMA packet backend without the native collective executor's memory or reduction costs. Both OptCC and the ordinary reduce-scatter/all-gather ring use this same network-only runner.

We reconstruct payload sizes from the disclosed section-size formulas; the JSON itself contains dependencies but no tensor or message bytes. Both schedules transmit $2(p-1)N$ data bytes in aggregate in the tested $L = 2$ cases. OptCC synchronization nodes carry an additional 20-byte control payload each. This validates byte accounting and schedule execution, not numerical AllReduce correctness. All receives are posted before roots launch, and each dependency is released at its source-side send completion or destination-side ACK-complete receive notification.

Transport gates exposed missing or premature whole-flow send callbacks at packet-size boundaries. We repaired final-packet serialization notification in the shared backend and repeated all 14 transport gates. The replay uses chronological serial event processing: the original multithreaded scheduler drained the OptCC graph but violated dependency timestamps even with one worker thread. Three scheduler micrographs, the 132-node $L = 2$ graph (255 edges), the 157-node $L = 1$ bubble graph, and a ring smoke test subsequently passed completion and trace checks. The earlier failed attempts are retained.

Additional deterministic checks. Within this declared setting, the OptCC schedule reduces network makespan by 10.4–28.8% relative to the matched ring. Four healthy-ring references and two preregistered segmentation checks also completed, bringing the measurement denominator to 30/30. At $p = 16$, $N = 15$ MiB, increasing k from 4 to 8 or 16 changed makespan from 1.986 ms to 2.143 ms or 2.886 ms; the primary $k = 4$ arm is retained. These measurements establish an executable comparator under bandwidth asymmetry. They do not measure route switching, packet-loss detection, recovery overhead, or end-to-end training throughput.

E Execution and CPU Emulation Records

The framed-TCP and native pending-flow targets have different execution contracts. These records retain their original integrity checks, failures and timing scopes; none proves exactly-once numerical AllReduce.

E.1 Framed-TCP Ownership Rules

Frames identify the round, step and route version and carry a payload length and checksum. Workers reject invalid frames and discard already consumed duplicates. The coordinator changes only the affected route slots, while interrupted steps retain completed send or receive work where the route is unchanged.

An in-flight receive has one owner per round and step. If a send-side abort interrupts the step while the receive route is unchanged, redo reuses the existing reader, including its partial frame and result. Creating another reader could consume the expected frame in an abandoned attempt. If the receive route changes, the superseded reader must stop before replacement. Deterministic tests cover consumed frames and partial headers across this abort; topology runs separately check completion, plan agreement and integrity. These checks do not constitute an exactly-once numerical AllReduce proof.

E.2 Recovery Outcomes and Native Transaction Cases

Fresh hard-fault, gray rate-degradation and transient checks each complete three lifecycles, with framed-transport integrity, schedule conformance and route-version agreement. All three hard-fault and all three sustained gray-degradation runs commit a localized route plan. Only two of the three transient runs suppress recovery; the remaining transient run commits after 1086.5 ms. Completion and transport integrity therefore do not establish correct transient discrimination. These nine runs also do not establish reliability for random loss.

The first original live-policy switch and stay attempts at 25% loss both select stay with unknown alternate spare; both subsequently exceed the 120-s round wait. The other 27 queued repetitions are explicitly unrun after the branch-gate failure. The guarded switch pilot completes 11 collective rounds (44 worker rounds), with only rank 2's three A slots changed and zero checksum or version errors. The guarded stay pilot selects its expected branch but also times out. Its failure blocks expansion to a repeated reliability claim. A policy label alone is not evidence that the selected arm minimizes completion cost.

Scripted pending-flow simulation gate. A separate native SimAI target tests runtime route changes before online detector integration. At launch it selects the committed plane, posts one matching receive and retains that transfer's ports and version. A shared, reliable modeled management FIFO charges 64 bytes per PREPARE, READY and COMMIT, with 1-Gb/s serialization and 1-μs propagation by default. Activation follows readiness and commit delivery at all four ranks. Management messages are not packetized on the RDMA data network.

All nine prospectively registered engineering cases pass: eight workloads complete with transport-map drain, and one deliberately nonconsecutive version is rejected. Independent auditing reconstructs all 582 trace records and verifies frozen inputs, source, binary and 39 shared libraries. The default transaction requested at 5 μs activates at 13.145 μs; two pending affected ring transfers move while launched transfers retain ownership. Two consecutive versions move four pending transfers. With 250-μs propagation, activation occurs at 759.609 μs after workload completion at 221.994 μs, so no transfer moves. A missing READY prevents activation and reaches its registered timeout at 1005 μs. Source-side and destination-side dependency chains and an unequal-size join also drain correctly. The equal-rate, impairment-free ring completes in 221.994 μs without control and 222.678 μs with a switch. These scripted checks establish pending-flow execution under the modeled protocol, not detection efficacy, beneficial recovery, cancellation/replay or numerical reduction. A severely stalled launched transfer cannot be rescued by this mode.

F Reference Policies and ACK-Progress Development

The original headroom heuristic below belongs to the CPU prototype. The remaining-work heuristic in Section 4 belongs to the separate native online loop. Their results are not interchangeable.

F.1 Earlier Headroom Policy

The current heuristic compares the affected plane's calibrated demand with the alternate plane's estimated spare service rate. A burst median and busy fraction provide demand and spare proxies; the rule switches when spare multiplied by a fixed reuse factor of 1.25 reaches the pre-fault demand. The factor is retained rather than tuned from the new outcomes. This rule is a heuristic, not a fitted model of total episode completion cost.

The original live estimate becomes unknown when its 256-sample window contains no bursts. A fault-induced collective pause can cause that condition even on a reachable alternate link. An opt-in guarded variant keeps the healthy spare estimate as a ceiling, takes a lower known live estimate, and uses the healthy reference when the live estimate is unknown. An unknown healthy reference still selects stay. This variant assumes stationary alternate capacity and cannot detect a silent capacity decrease on an idle port. Both inputs and the assumption are logged.

Native causal ACK-progress development. A separate native target samples sender acknowledgment progress every 2 ms, combining completed flow bytes with active queue-pair ACK credit. The rule calibrates the median of four positive fully busy service samples, then alarms after three busy samples below half that reference. Idle windows interrupt evidence. Reports occupy 64 bytes each on a separate reliable 10-Mb/s FIFO with 1-ms propagation. The detector runs on delivered reports inside simulator callbacks; injection metadata is excluded. This signal concerns ACK progress, not permanent-loss proof.

Four deterministic development scenes execute three consecutive four-rank 3-MiB rings with 1-Gb/s accesses. Persistent rank-2 A service reduction to 0.2 Gb/s at 50 ms alarms after 9.1536 ms; healthy and fully restored 2-ms reduction scenes produce no alarms. Changing report propagation to 120 ms delays the alarm to 128.1536 ms, failing the deadline. An earlier device-only restoration left a newly

created queue pair capped at the reduced rate for its 31.563-ms transfer and produced a registered control alarm. That ledger remains intact. The amended injector changes directed configured NIC service and active DCQCN pacing caps at both reduction and restoration; detector logic is unchanged. This model correction is not an algorithm-only gain or a hidden physical-link fault simulation. Independent replay verifies all 2920 delivered reports and 226492416 ACK bytes in the amended four-case ledger. These are development cases, not broad held-out validation of the primary constraint.

G Additional Cost and Scaling Diagnostics

G.1 Static Calibration Setup

The healthy 200/200 reference completes in 0.222 ms, and a 200/20 reference completes in 1.128 ms. At 40/200, switching reduces makespan by 22.9%; at 40/20 it increases makespan by 33.0%. These are static network comparisons, not LIMER controller results. Three initial 40/20-Gb/s attempts aborted before simulated time advanced because the default ECN tables omitted those link speeds. We retained them, added explicit thresholds scaled from the supplied 25-Gb/s row and repeated all six calibrations; the two constrained-B references were separately registered. Runtime delay, background demand and finite-horizon costs remain necessary for the route-choice study.

G.2 Severe-Loss Fixed-Route Contrast

A separate fixed-route diagnostic retains the framed-TCP workload: 25% random loss on rank-2 A, 10-Mb/s rank-2 B access, and eight post-fault rounds. All three fixed-stay attempts exceed the 120-s round wait; they complete zero, one and zero post-fault rounds, respectively. All three immediate safe-boundary localized-switch attempts complete all eight rounds with integrity and locality checks, in 84.804–85.029 s from fault injection to the last round's completion. The switch changes only rank 2's three A slots. Source and container image are shared, but realized packet-loss draws are not paired.

This diagnostic omits detection delay and in-flight cutover by moving at the next safe round. It cannot establish adaptive LIMER superiority or quantify the counterfactual completion time of a timed-out arm. It does show why insufficient estimated spare is not an optimality label: staying on a highly impaired TCP path can fail to finish even when the alternative is slow. Combined with the static calibration's opposite preferred arms, the result motivates a finite-episode cost objective that distinguishes bandwidth reduction from random loss.

G.3 Scaling with SimAI

The current baseline includes 5- and 16-rank single-plane graphs; the dual-plane calibration includes four logical ranks. The planned crossed schedule/route matrix and expansion up to 128 ranks remain unrun. At unchanged 10-ms two-ended reporting, the modeled collector load is 204.8 Kb/s per logical rank: eight ranks exceed the current 1-Mb/s service rate, and 128 ranks would offer 26.2144 Mb/s. This is an algebraic budget implication, not a measured 128-rank result. Simulated control delays and actual implementation latencies remain separate.

H Implementation Environments

H.1 New RDMA Loop and Reproducibility

The 2026-10-28 native backend is compiled with GCC 12.2.0 in an isolated Linux container. It is a CPU packet simulation; a logical GPU is a modeled endpoint. Each campaign retains its exact registration, generated topology and fault inputs, source and native-binary hashes, frozen scorer, raw event archive and attempt ledger. Independent replay reconstructs event evidence; full archive digests and compact evidence are checked separately. Native validation uses 140 unit tests at its execution checkpoint; a later report portability review passes 142 tests after adding two rounding-guard tests. That review changes only postprocessing: a sub-picosecond tolerance for aggregate mean roundoff across Python versions. Per-trial, pair, count and p95 checks remain exact; native rules, seeds and registered evidence are unchanged. The new report regenerates the same result JSON.

H.2 Historical Component Environments

The following environment and test counts describe their recorded execution checkpoint, not a newly run verification or a current full-suite count.

The CPU harness uses Mininet, kernel Open vSwitch and a four-worker framed TCP ring. Each worker has a persistent outgoing and incoming connection on each plane. Faults act on a directed worker egress, while the detector observes the switch-facing peer. Workers stream step events and plan versions to the coordinator; results retain per-round byte counts, fingerprints, checksums, route changes and failures.

Fresh execution uses isolated rootless Docker containers on an approved Linux host, Ubuntu 24.04/Python 3.12 and a nominal 4-CPU/4-GiB budget per lifecycle, with at most two concurrent Mininet runs. Namespace and sysfs mounting require privileged mode inside the rootless daemon. Resource limit warnings and actual sampling intervals are retained. The complete Linux unit suite passes 292 tests on the Linux Python 3.12 host. These checks establish implementation behavior, not deployment on GPU collectives or switch hardware.

The separate SimAI adapter uses the pinned network frontend and packet backend. The single-plane comparator reconstructs graph byte sizes and endpoint dependency releases. The dual-plane calibration adds two physical network endpoints per logical rank, with disjoint switches and plane-consistent sends and receives. It changes static slot placement before launch and includes no runtime detector or cutover delay. A separate dynamic target registers a single matching receive at each launch and retains its physical endpoints and version through completion. Only pending affected transfers can change planes; launched transfers remain on their original route.